Chakra Orb Hand Tracker (Python)

Open your palm in front of your webcam and a glowing orb forms above your
hand, using real-time hand tracking. Close your palm and it fades away.
Built with Python, OpenCV, and MediaPipe.

How it works

1. **MediaPipe Hands** detects 21 landmark points per hand from the webcam
   feed (wrist, knuckles, fingertips).
2. **Open-palm detection**: for each finger, the code compares how far the
   fingertip is from the wrist versus how far the knuckle (PIP joint) is
   from the wrist. If a fingertip is farther out, that finger counts as
   "extended." Three or more extended fingers = open palm.
3. **Charge system**: while the palm stays open, a `charge` value (0 to 1)
   increases each frame; while closed, it decreases.

This makes the orb
grow and shrink smoothly instead of popping in and out.

4. **Drawing the orb**: OpenCV draws several semi-transparent circles of
decreasing size (outer glow → mid glow → bright core) blended with
`cv2.addWeighted`, plus three rotating lines for a spinning effect.

Setup

```
pip install -r requirements.txt
python rasengan.py
```

Press `q` to quit.

Requirements

- Python 3.8+
- A webcam
- Packages: `opencv-python`, `mediapipe`, `numpy` (see `requirements.txt`)

Project structure

```
.
├── rasengan.py       # main script: hand
```

```
tracking + orb rendering
├── requirements.txt  # dependencies
└── README.md
```

Customization

Open `rasengan.py` and tweak:

Setting	Variable	What it does
Orb grow speed	`CHARGE_UP_SPEED`	Higher = orb forms faster
Orb fade speed	`CHARGE_DOWN_SPEED`	Higher = orb disappears faster
Open-palm sensitivity	`extended >= 3` in `is_palm_open`	Lower = easier to trigger
Orb colors	`PALETTES` dict	Change glow/core colors per hand
Orb height above	`1.3` in `orb_center` calculation	Higher = orb floats

Setting	Variable	What it does
palm		farther up

Credits

Concept inspired by chakra-orb hand-tracking demos built with MediaPipe;
this implementation is written in Python using OpenCV instead of the
browser/JavaScript versions of the idea.